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Impact of therapeutic exercise on craniocervical angle in forward head posture: a systematic review and meta-analysis

Andoni Carrasco-Uribarren ^{a,b}, Luis Ceballos-Laita ^b, Silvia Pérez-Guillén ^a, Sandra Jiménez-Del-Barrio ^b, Daniel Pantaleón-Hernández ^a and Sara Cabanillas-Barea ^a

^aDepartamento de Fisioterapia, Facultad de Medicina y Ciencias de la Salud, Universidad Internacional de Cataluña, Barcelona, España;

^bGrupo de Investigación Clínica en Ciencias de la Salud, Departamento de Cirugía, Universidad de Valladolid Soria, España

ABSTRACT

Introduction: Forward head posture (FHP) is one of the most frequently reported postural deviations in contemporary populations. Therapeutic exercise has been associated with improvements in FHP, rounded shoulders, and thoracic kyphosis. However, it remains unclear whether exercise programs targeting the neck region differ in their effects from those that also include the thoracic spine compared with control conditions.

Methods: A systematic review with meta-analysis was conducted following a comprehensive search of the Cochrane Library, PubMed, PEDro, and Web of Science databases from inception to 9 February 2026. Eligible studies were randomized controlled trials investigating therapeutic exercise interventions in patients with FHP. The intervention had to be based on therapeutic exercise. Studies were required to compare the intervention with a control group. Methodological quality was assessed using the PEDro scale, and the certainty of the evidence was evaluated according to the GRADE guidelines.

Results: Thirteen randomized controlled trials involving 819 participants were included in qualitative synthesis and meta-analysis. Both neck-focused exercise and combined neck – thoracic exercise programs showed statistically significant short-term improvements in craniocervical angle and neck disability compared with control conditions. Combined neck – thoracic exercise programs demonstrated significant reductions in pain intensity post-intervention. The certainty of the evidence ranged from low to very low.

Conclusion: Therapeutic exercise programs targeting the neck region, alone or combined with thoracic exercises, may improve craniocervical angle and neck-related outcomes in individuals with FHP. However, the certainty of the evidence ranged from low to very low and the findings should be interpreted with caution.

Protocol registration: PROSPERO registration ID CRD420261304664.

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KEYWORDS

Exercise; forward head posture; systematic review; meta-analysis

Background

Globally, neck pain affects approximately 2,450 individuals per 100,000 population, representing a substantial public health burden [1]. Neck pain is considered a multifactorial condition, and forward head posture (FHP) has been proposed as one of the factors to be considered, with several studies reporting an association between neck pain and FHP [2,3]. FHP is one of the most frequently reported postural deviations in contemporary populations. Prevalence estimates vary considerably depending on the occupational group and the diagnostic criteria applied, ranging from approximately 11% to 60% across different populations [4,5]. However, this relationship remains controversial, as the available evidence appears to be age-dependent and based on cross-sectional studies, precluding conclusions about causality [6].

FHP is commonly defined as an anterior translation of the head relative to the cervical spine. It has been

hypothesized to reflect a pattern of muscular imbalance characterized by shortening of the pectoralis major, upper trapezius, and levator scapulae, along with weakness of the deep cervical flexors, lower trapezius, and rhomboids [7,8]. The craniocervical angle (CVA) has been widely used to quantify forward head translation [9]. The CVA angle is formed by the intersection of a horizontal line passing through the spinous process of C7 and a line connecting C7 to the tragus of the ear [9]. Smaller CVA values indicate greater anterior head translation. While different thresholds have been reported in the literature, a CVA smaller than 48°–50° is one of the most commonly adopted cutoff to classify individuals as presenting FHP [10]. Previous literature has also suggested that CVA values between 30° and 50° may represent mild-to-moderate FHP, whereas values below 30° may indicate severe FHP [11].

Various physiotherapy interventions have been investigated to address FHP, with findings suggesting

that both passive and active approaches may contribute to improvements in postural alignment, pain intensity, and disability [12–14]. The combination of postural corrective exercises and manual therapy interventions has been associated with improvements in pain and functional outcomes in patients with FHP [14]. When therapeutic exercise alone has been analyzed, improvements in FHP, in rounded shoulders and in thoracic kyphosis have been described. Nevertheless, most available evidence derives from studies in individuals with upper crossed syndrome and from multimodal corrective exercise programs addressing multiple postural components simultaneously [12,13]. However, many of these studies did not include deep cervical flexor training, despite its association with FHP [7,8] and evidence suggesting superior effects on posture, pain, and disability compared with other exercise modalities [15].

Several reviews have suggested that exercise interventions, either alone or combined with manual therapy, may improve posture in individuals with FHP [12,16–18]. However, to the best of the authors' knowledge, no previous review has adopted a CVA < 50° as an inclusion criterion, although this threshold is the most frequently used to define FHP. In addition, it remains uncertain whether exercise interventions specifically targeting the neck region differ in their effects from programs that also incorporate the thoracic spine. Therefore, the aim of this review was to systematically evaluate the effectiveness of exercise-based interventions for FHP, applying a CVA smaller than 50° as the inclusion criterion, and to compare (1) neck-focused exercise programs versus control conditions and (2) combined neck and thoracic exercise programs versus control conditions.

Methods

Study design

The present review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and Cochrane recommendations [19]. The protocol was prospectively registered in PROSPERO (registration number: CRD420261304664).

Search strategy

The Physiotherapy Evidence Database (PEDro), PubMed (MEDLINE), the Cochrane Library, and Web of Science (WoS) were searched from inception to 9 February 2026. The search strategy was developed using gray terms and Medical Subject Headings (MeSH). A detailed description of the search strategy for each database is provided in Supplementary material.

Eligibility criteria

Eligibility criteria were defined according to the PICOS framework as follows. Population: Patients with FHP, operationally defined as a CVA < 50°; Intervention: Therapeutic exercise – based interventions that included at least one exercise specifically targeting activation of the deep neck muscles; Comparison: Control condition (e.g. no intervention, usual care); Outcomes: Primary CVA, and secondary neck disability and pain intensity; Study design: randomized controlled trials.

Studies were excluded if participants did not meet the predefined criterion of a CVA < 50° (when the CVA threshold was not explicitly stated in the inclusion criteria, baseline descriptive data were reviewed to verify that the study sample fulfilled this requirement), if the article was not published in English, if the intervention did not incorporate at least one exercise specifically targeting activation of the deep cervical musculature, or if the reported outcomes were not aligned with the research objectives or were assessed using instruments without established validity or reliability.

Study selection and data extraction

Two independent reviewers (A.C.-U. and L.C.-L.) screened the retrieved records. Titles and abstracts were assessed to determine potential eligibility. Full-text articles were then evaluated for studies meeting the inclusion criteria. Any discrepancies between reviewers were resolved by a third reviewer (S.C.-B.).

The same two reviewers independently performed data extraction using a predefined data extraction form developed in accordance with Cochrane Collaboration guidelines [19]. Extracted data included samples characteristics, details of the intervention, outcome measures, and main results.

Methodological quality assessment

Methodological quality was independently assessed by two reviewers (A.C.-U. and L.C.-L.) using the PEDro scale [20]. The PEDro scale comprises 11 items evaluating the internal validity and statistical reporting of randomized controlled trials. As the first item relates to eligibility criteria and external validity and is not included in the total score, the maximum attainable score is 10. Studies were classified as high quality if they scored ≥ 7, fair quality if they scored 5–6, and poor quality if they scored ≤ 4 [21].

Data synthesis and analysis

Data were synthesized qualitatively and, whenever possible, quantitatively.

The combination of data was performed when at least two studies were comparable using RevMan 5.4 software. The studies were considered comparable when therapeutic exercise targeted the neck region alone or a combination of the neck and thoracic regions, and when a common comparator was used. The mean change scores, the standard deviation (SD) or the 95% confidence interval (CI), and the sample sizes for each group were extracted or calculated. The mean difference (MD) and standard deviation (SD) from the post-intervention data were used when change scores were unavailable or could not be calculated, following the recommendations of the Cochrane Handbook for Systematic Reviews of Interventions [22].

A random-effects model was used to perform the meta-analysis to account for the possibility that the included studies were not estimating the same underlying intervention effect [23]. Heterogeneity was assessed by evaluating the similarity of point estimates, the overlap of confidence intervals, the study context, and the I^2 statistic derived from the forest plots [24,25]. To assess potential publication bias and evaluate the influence of individual studies, we visually inspected the forest plots and conducted sensitivity analyses by sequentially excluding individual studies. Funnel plots were not performed because none of the meta-analyses included ≥ 10 studies, which is the recommended minimum threshold for their use.

Certainty of evidence assessment

The certainty of the evidence was independently assessed by two reviewers using the GRADE approach and summarized in a GRADE Evidence Profile. The certainty evidence dimensions were evaluated according to GRADE methodology. The quality of evidence was downgraded based on the following factors: risk of bias, inconsistency, indirectness, imprecision, and publication bias. The overall certainty of evidence was classified as high, moderate, low, or very low according to these domains, supporting interpretation of the strength and reliability of the findings [25].

Results

After the search strategy, 722 records were identified. After removing duplicates and screening titles and abstracts, 37 RCTs were selected for full-text review. Thirteen RCTs were included in the qualitative and quantitative analyses. Twelve studies were excluded because they did not use an appropriate comparator to allow extrapolation of the effects of neck exercise or combined neck and thoracic exercise [26–37]. Three studies were excluded because they did not include activation of the deep neck flexors as part of the intervention [38–40]. Six studies were excluded because they did not use the CVA to measure FHP [41–46]. Finally, three studies were excluded because

participants presented baseline CVA values greater than 50° [47–49]. In the PRISMA flowchart the selection process is shown (Figure 1).

Characteristics of the included studies

Thirteen RCTs were included, involving a total of 819 patients with FHP [54,55–61]. Eight studies required a CVA of less than 50° as an inclusion criterion for FHP [2,50,51,53,55–57,59], while in the remaining studies this cutoff was not predefined; however, baseline CVA mean values were still below 50° [52,54,58,60,61], indicating that most samples exhibited clinically relevant FHP. The most frequently employed control condition was no intervention, representing 77% of the included studies [2,52–60]. Alternative control interventions included muscle energy technique [50], conventional treatment consisting of TENS and isometric neck exercises [51], and self-myofascial release combined with stretching [61]. Intervention duration ranged from 4 to 10 weeks, with three sessions per week being the most prescribed training frequency.

The characteristics of the included studies, including publication year, participant demographic characteristics and diagnosis, type of intervention and comparator details, follow-up duration, outcome measures, and main results, are summarized in Table 1. Table 2 presents more details of the intervention and both groups intervention details.

Methodological quality

Considering the scores of the PEDro scale, five studies [2,50,53,57,59] were considered as high quality, five studies [52,55,56,58,61] as a fair quality, and three studies [51,54,60] as a poor quality. Table 3 showed the PEDro scale.

Synthesis of results

Craniocervical angle

The very low certainty of the evidence suggests that 4–6 weeks of neck exercise may result in a statistically significant change compared with a control intervention (MD = -2.63 ; 95%CI $-4.50, -0.75$; 3 studies, 163 patients) [2,51,55]. The very low certainty of the evidence (downgraded for risk of bias, inconsistency and imprecision) suggested that 8 weeks of neck exercise showed no statistically significant change compared with a control (MD = -6.62 , 95%CI $-13.94, 0.70$; 2 studies, 100 patients) [2,52] (Figure 2(A)).

The very low certainty of the evidence suggests that 4–6 weeks of neck + thoracic exercise may result in a statistically significant change compared with a control intervention (MD = -3.71 ; 95%CI $-7.37, -0.05$; 3 studies, 103 patients) [50,56,61]. The very low

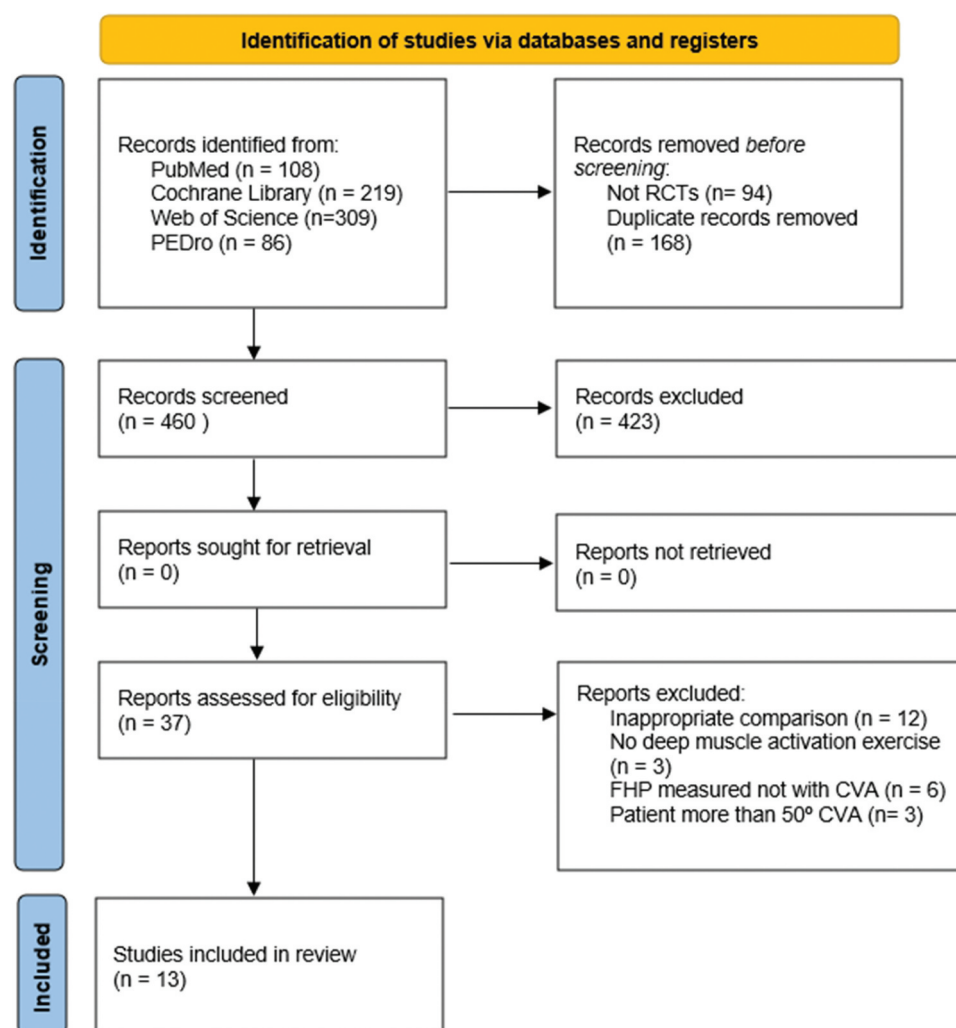


Figure 1. Flowchart diagram.

certainty of the evidence (downgraded for risk of bias, inconsistency and imprecision) suggested that 8–10 weeks of neck+thoracic exercise showed statistically significant change compared with a control (MD = −4.15, 95%CI −5.52, −2.78; 6 studies, 256 patients) [53,54,57–60](Figure 2(B)).

Neck disability

Low certainty of the evidence suggests that 4 weeks of neck exercise results in a statistically significant difference compared with a control intervention (MD = −3.13; 95%CI −4.25, −2.00; 2 study, 110 patients) [2,51]. The very low certainty of the evidence suggests that 8 weeks of neck exercise results in a statistically significant difference compared with a control intervention (MD = −7.19; 95%CI −8.26, −6.12; 1 study, 70 patients) [2](Figure 2(C)).

The very low certainty of the evidence suggests that 4 weeks of neck + thoracic exercise results in a statistically significant difference compared with a control intervention (MD = −12.16; 95%CI −16.02, −8.30; 1 study, 40 patients) [50]. The very low certainty of the evidence suggests that 8 weeks of neck + thoracic exercise showed no statistically significant

difference compared with a control intervention (MD = −11.20; 95%CI −26.71, 4.31; 2 studies, 108 patients) [57,60](Figure 2(D)).

Pain intensity

The very low certainty of the evidence suggests that 4 weeks of neck + thoracic exercise may result in a statistically significant change compared with a control intervention (MD = −4.85; 95%CI −5.78, −3.92; 1 study, 40 patients) [50]. The very low certainty of the evidence suggests that 8 weeks of neck + thoracic exercise results in a statistically significant change compared with a control intervention (MD = −2.62; 95%CI −3.79, −1.45; 4 studies, 180 patients) [53,57,58,60](Figure 2(E)).

The certainty of the evidence for CVA, neck disability, and pain intensity was downgraded due to risks of bias, inconsistency, and imprecision. Table 4 showed the GRADE

Discussion

FHP is a prevalent postural disorder associated with neck pain and disability. Although therapeutic exercise

Table 1. Characteristics of the studies.

Author	Participants		Intervention		Duration between treatment and follow-up	Outcome (tool)	Main results
	N (sex ratio)	Mean age (SD)	Diagnosis	IG			
[50]	40	IG: 42.9 ± 8.5 CG: 44.4 ± 10.1	UCS FHP	Multimodal group (n = 20)	Control (n = 20)	Post-treatment 4 weeks	CVA ↑ VAS ↑ NDI
[51]	40 (12 M/28 F)	IG: 29.9 ± 6.0 CG: 32.6 ± 10.6	CNP FHP	CST + Conventional treatment	Conventional treatment (n = 20)	Post-treatment 4 weeks	CVA ↑ NPRS ↑ NDI
[52]	30 (4 M/26 F)	IG: 21.2 ± 5.1 CG: 20.7 ± 1.7	TMJ FHP	Standard treatment + CST (n = 15)	Standard treatment Advice (n = 20)	Post-treatment 8 weeks	ND between-groups CVA
[53]	60	24.7 ± 2.1	NSCNP FHP	Therapeutic exercise (n = 20)	Control (n = 30)	Post-treatment 8 weeks	↑ CVA ↑ VAS ↑ CVA ↑ VAS
[60]	60 (19 M/41 F)	IG: 22.0 ± 2.7 CG: 20.8 ± 1.7	FHP	Therapeutic corrective exercise (n = 30)	Control (n = 30)	Post-treatment 8 weeks	↑ VAS ↑ NDI
[54]	41 (15 M/26 F)	IG: 38.4 ± 11.3 CG: 36.9 ± 9.5	FHP	Therapeutic exercise (n = 27)	Control (n = 17)	Post-treatment 10 weeks	ND between group in CVA
[55]	60	IG: 22.0 ± 1.2 CG: 21.6 ± 1.5	FHP	Therapeutic exercise (n = 30)	Control (n = 30)	Post-treatment 6 weeks	↑ CVA
[57]	48 (48 F)	IG: 34.2 ± 2.7 CG: 34.6 ± 3.3	CNP FHP	Therapeutic exercise (n = 24)	Control (n = 24)	Post-treatment 8 weeks	↑ VAS ↑ NDI ↑ CVA
[56]	30	IG: 22.6 ± 2.9 CG: 21.2 ± 1.3	RF FHP	McKenzie exercise (n = 15)	Control (n = 15)	Post-treatment 4 weeks	↑ CVA
[58]	32 (32 F)	IG: 37.7 ± 3.8 CG: 35.5 ± 5.4	CNP FHP	Therapeutic exercise (n = 16)	Control (n = 16)	Post-treatment 8 weeks	↑ VAS ↑ CVA
[2]	70 (44 M/26 F)	IG: 48.6 ± 4.1 CG: 48.5 ± 5.7	Rad. FHP	Biofeedback therapeutic exercise (n = 35)	Control (n = 35)	Post-treatment 4 weeks	↑ NDI
[59]	36 (19 M/17 F)	IG: 23.0 ± 0.8 CG: 23.08 ± 1.8	FHP	Therapeutic exercise (n = 16)	Control (n = 16)	Post-treatment 8 weeks	↑ CVA ↑ NDI
[61]	80	IG: 19.5 ± 1.4 CG: 20.2 ± 2.0	FHP	Self-Myofascial Release + Stretching + Strengthening Group (n = 20)	Self-Myofascial Release + Stretching Group (n = 20)	Post-treatment 4 weeks	CVA ND between group

SD: Standard deviation; IG: Intervention group; CG: Control group; UCS: Upper Crossed Syndrome; FHP: Forward head posture; CVA: Cervical angle; VAS: Visual analog scale; NDI: Neck disability index; M: Male; F: Female; CNP: Chronic neck pain; CST: Cervical stabilization exercises; NRS: Numeric rating scale; TMJ: Temporomandibular joint; ND: No difference; NSCNP: Nonspecific chronic neck pain; RF: Respiratory function; Rad: Radiculopathy.

Table 2. Characteristics of the intervention and control.

Author	Experimental Intervention Description	Exercise Components	Frequency (S/W)	Program duration (W)	Session duration (min)	Control
[50]	Multimodal approach	MET + cervical and scapular stabilization + education	3	4	Not Reported	MET only
[51]	Cervical stabilization exercises + conventional	Craniocervical flexion+TENS+cervical Isometric	3	4	Not Reported	TENS+ Cervical Isometric
[52]	Cervical stabilization training	Cervical + scapular strengthening	3	6	Not Reported	No intervention
[53]	Therapeutic exercise routine	Cervical exercises	3	8	Not Reported	No intervention
[60]	Corrective Exercises Protocol	Cervical + scapular strengthening	3	6	Not Reported	No intervention
[54]	Strengthening program	Cervical + scapular strengthening	4	10	30	No intervention
[55]	Exercise + sensorimotor training	Cervical motor control + proprioception	Daily	4	Not Reported	No intervention
[57]	Therapeutic exercises routine	Cervical + scapular strengthening	3	8	20–30	No intervention + Advise
[56]	McKenzie cervical exercises	Cervical strengthening	3	4	20	No intervention
[58]	Corrective strengthening protocol	Cervical and scapular exercise + education	3	8	30	No intervention + advise
[2]	Cranio-cervical flexion training with biofeedback	Deep cervical flexors	3	8	13	No intervention
[59]	Corrective exercise program	Deep cervical flexors + scapular strengthening + stretching	3	8	20	No intervention
[61]	SMR+ stretching+ Strengthening	Cervical+ Scapular SMR + Stretching+ Strengthening	3	4	Not Reported	Cervical+ Scapular SMR + Stretching

S/W: session/week; W: week; min: minute; MET: Muscle energy technique; No int.: No intervention; SMR: Self-Myofascial Release.

Table 3. Pedro scale scores.

Autor	Items											Total
	1	2	3	4	5	6	7	8	9	10	11	
Aneis YM. et al.,	Y	Y	Y	Y	N	N	Y	Y	N	Y	Y	7/10
Arif T. et al.,	N	Y	N	Y	N	N	N	Y	N	N	N	3/10
Arslan M. et al.,	Y	Y	Y	Y	N	N	Y	N	N	Y	Y	6/10
Dareh-Deh HR. et al.,	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Y	8/10
F. Sarraf et al.,	Y	Y	N	Y	N	N	N	Y	N	N	Y	4/10
Harman K. et al.,	Y	Y	N	Y	N	N	N	N	N	Y	Y	4/10
Jantoon N. et al.,	Y	Y	Y	Y	N	N	Y	N	N	Y	Y	6/10
Letafatkar A. et al.,	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Y	8/10
Kim SY. et al.,	Y	Y	N	Y	N	N	N	Y	N	Y	Y	5/10
Mehri A. et al.,	Y	Y	N	Y	N	N	Y	N	N	Y	Y	5/10
Mohamed A. et al.,	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Y	8/10
Sarıyıldız A. et al.,	Y	Y	N	Y	N	N	Y	Y	N	Y	Y	7/10
Titcomb DA. et al.,	Y	Y	Y	Y	N	N	Y	N	N	Y	Y	6/10

1, eligibility criteria 2, random allocation; 3, concealed allocation; 4, similarity at baseline; 5, blinding of participants; 6, blinding of therapists; 7, blinding of assessors; 8, measures of at least one key outcome from at least 85% of participants initially allocated to groups; 9, intention to treat analysis; 10, between-group comparison; 11, point measures and measures of variability. 1 = Yes (1 point), 0 = No (0 point), maximum score = 10 (criterion 1 is not included in scores).

has been shown to improve FHP, its effects in individuals defined by a specific CVA threshold ($<50^\circ$) have not been specifically investigated. Therefore, this systematic review and meta-analysis aimed to evaluate the effectiveness of exercise-based interventions and to compare neck-focused and combined neck – thoracic exercise programs versus control conditions in individuals with FHP defined by a CVA $<50^\circ$.

According to the eligibility criteria thirteen studies were included in the present systematic review and meta-analysis. The results demonstrated statistically

significant short-term effects of both exercise approaches on CVA at 4–6 weeks of follow-up. However, these findings should be interpreted with caution, as the certainty of the evidence was rated as very low, limiting confidence in the magnitude and consistency of the results. At longer follow-up (8–10 weeks), only combined neck – thoracic exercise programs maintained statistically significant improvements in CVA. Although this may suggest a potential added benefit of including the thoracic region, the low certainty of the evidence precludes firm conclusions.

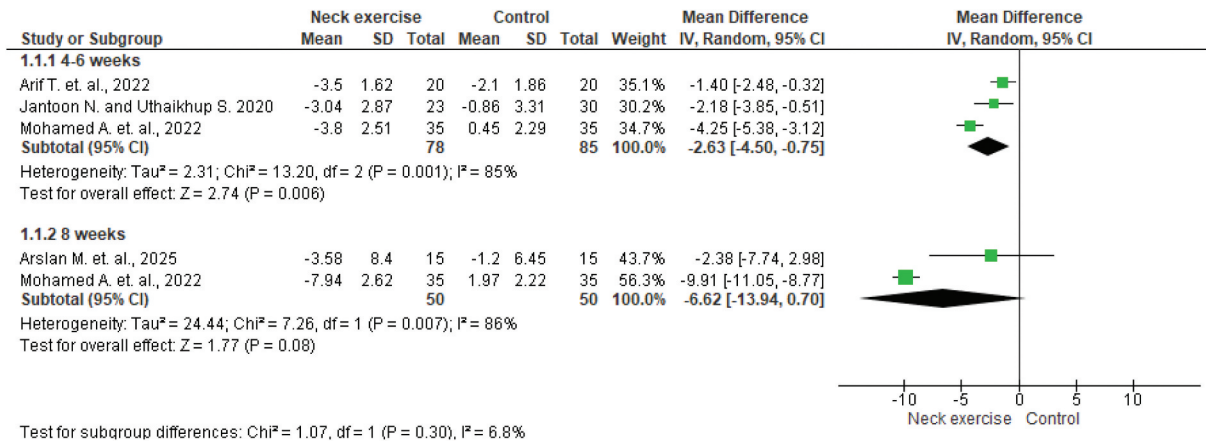


Figure 2A. Forest plot of the craniocervical angle of the neck exercise versus control.

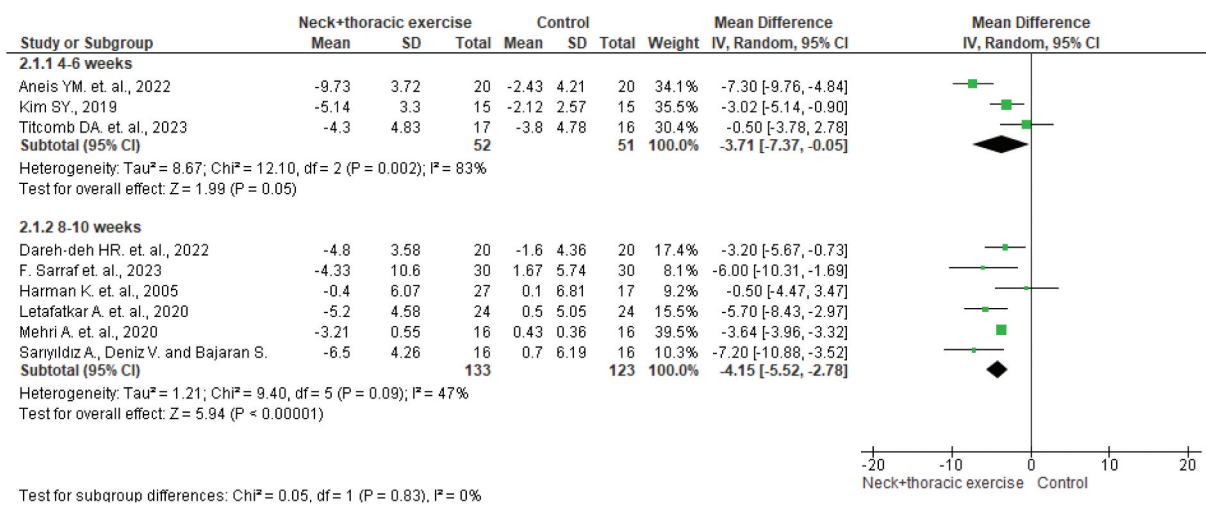


Figure 2B. Forest plot of the craniocervical angle of the neck+thoracic exercise versus control.

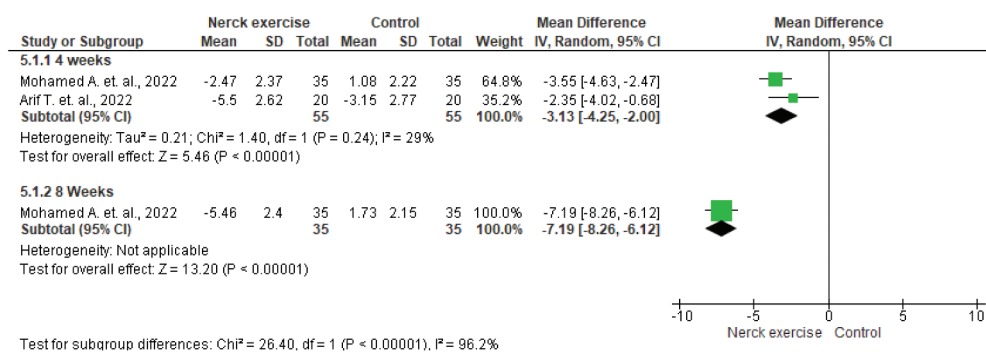


Figure 2C. Forest plot of the neck disability of the neck exercise versus control.

Regarding neck disability and pain intensity, the improvements observed should also be interpreted cautiously, given the limited number of studies and the overall low-to-very-low certainty of the evidence.

According to the PEDro scale, most of the included trials demonstrated high (38,4%), and fair (38,4%) methodological quality. Blinding of therapists and participants was not implemented in any study, which may increase the risk of performance bias; however, this limitation is common in clinical trials based on physiotherapy interventions [62]. Five studies failed

to meet item 8 of the PEDro scale, as outcome data for at least one key variable was not obtained from at least 85% of the participants initially allocated to each group. Finally, several RCTs did not report the use of an intention-to-treat analysis, especially in those trials where participant loss occurred, which may introduce bias and potentially overestimate treatment effects.

The results of this study are consistent with findings by other systematic reviews [12,17], which reported that therapeutic exercises are effective in improving FHP. In contrast to Sepehri et al. [12], we included

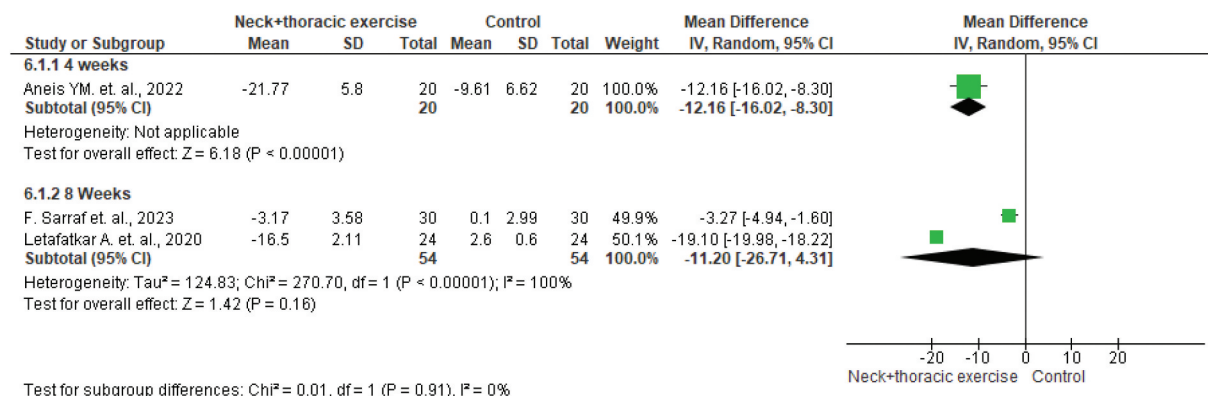


Figure 2D. Forest plot of the neck disability of the neck + thoracic exercise versus control.

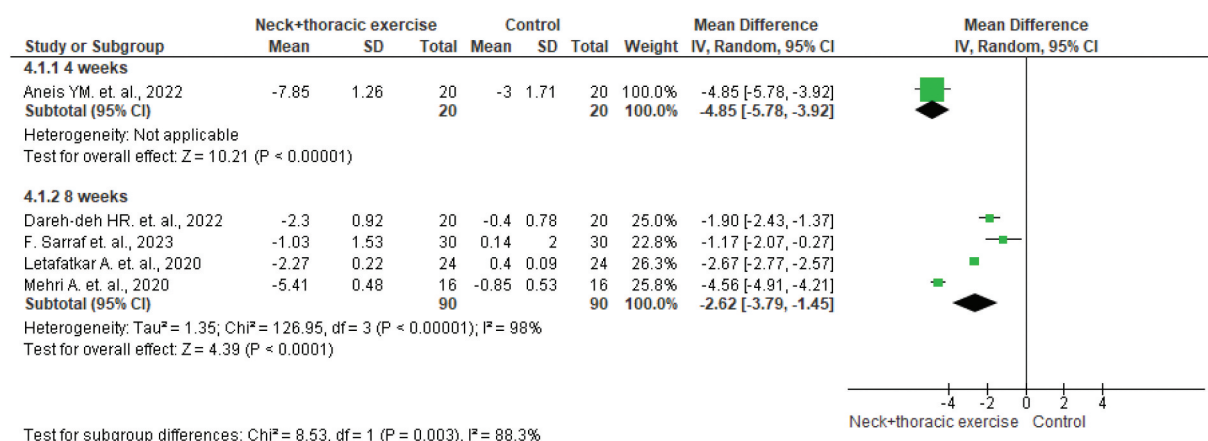


Figure 2E. Forest plot of the pain intensity of the neck+thoracic exercise versus control.

Table 4. Certainty of evidence according to GRADE profile.

Outcome	No. of studies (participants)	Risk of bias	Inconsistency	Indirectness	Imprecision	Certainty of evidence
CVA						
4–6 weeks	3 (163)	Serious ^b	Very serious ^c	Serious ^e	None	Very low
8 weeks	2 (100)	Serious ^b	Very serious ^c	Serious ^e	None	Very low
CVA neck + thoracic						
4–6 weeks	3 (103)	Very serious ^a	Very serious ^c	Serious ^e	None	Very low
8–10 weeks	6 (156)	Very serious ^a	None	Serious ^e	None	Very low
Neck disability						
4 weeks	2 (110)	serious ^b	None	Serious ^e	None	Low
Neck disability +thoracic						
8 weeks	2 (108)	Very serious ^a	Very serious ^c	Serious ^e	None	Very low
Pain intensity						
8 weeks	4 (180)	Very serious ^a	Very serious ^c	Serious ^e	None	Very low

Explanations.

a. Downgraded two levels for risk of bias (more than 50% of the patients were from studies with fair methodological quality)

b. Downgraded one levels for risk of bias (more than 25% of the patients were from studies with fair methodological quality)

c. Downgraded two level for inconsistency (heterogeneity of results indicated by $I^2 > 75\%$).

d. Downgraded two level for inconsistency (heterogeneity of results indicated by $I^2 > 50\%$).

e. Downgraded one level for for indirectness (downgraded by one level if different populations, interventions, or comparators were included).

f. Downgraded one level for imprecision (less than 100 patients per group).

High: We are very confident that the true effect is close to the estimate of the effect.

Moderate: We are moderately confidence in the effect estimate. The true effect is close to the estimate of the effect, but the result can be different.

Low: Confidence in the effect estimate is limited, the true effect can be substantially different from the estimate of the effect.

Very Low: There is little confidence in the effect estimate, the true effect is likely to be substantially different from the estimate effect.

studies that quantified FHP using CVA. Furthermore, we categorized the studies into those implementing neck exercise alone and those combining neck and thoracic exercises. The results showed that both neck exercise and thoracic exercise achieved improvements exceeding the minimal clinically important difference

(MCID) (1.40°) [63]. The MCID was established in young students with a CVA < 50°, which differs from the sample included in this systematic review; therefore, the findings should be interpreted with caution.

Although previous evidence has suggested that improving FHP may positively influence neck pain

intensity and disability, supporting the clinical relevance of postural correction strategies [14], the present review was not specifically designed to examine the association between postural changes and clinical outcomes. Moreover, evidence from related fields, such as the scapular dyskinesia, suggests that exercise may improve pain and disability despite limited or inconsistent biomechanical changes [64,65]. These findings highlight the complexity of the relationship between postural alignment and clinical outcomes and suggest that improvements in symptoms may not be solely explained by biomechanical changes.

Our results suggest that both neck exercise and combined neck – thoracic exercise programs improve neck disability compared with controls. However, although the neck – thoracic exercise program did not reach statistical significance for disability, the magnitude of improvement exceeded the MCID established for the NDI (5.5 points) [66]. Regarding pain intensity, a meta-analysis could only be performed for the neck – thoracic exercise program. This analysis demonstrated statistically significant reductions in pain, with improvements exceeding the 1.5-point MCID [66]. These findings are consistent with previous studies comparing exercise interventions with control groups in patients with chronic neck pain and forward head posture [16]. Similarly, the meta-analysis conducted by Rahman Sheikhhoseini et al., which included individuals with FHP, reported beneficial effects of therapeutic exercise on CVA and neck pain intensity [17]. Although these findings may be clinically relevant, existing evidence suggests that multimodal approaches combining exercise and manual therapy may provide greater benefits than exercise alone in individuals with neck pain [18,67].

Limitations

This systematic review has some limitations. Although a CVA < 50° was applied as an inclusion criterion in this review, not all included studies predefined this threshold in their eligibility criteria. In several cases, eligibility was inferred from baseline CVA mean values, which may introduce indirectness and compromise the homogeneity of the study samples. Follow-up periods were short (maximum 10 weeks), precluding conclusions about the long-term effects on postural, functional, and pain-related results. The small number of trials included in each meta-analysis and the limited sample sizes contributed to imprecision. Finally, although several major databases were searched, the exclusion of additional databases and non-English publications may have limited the comprehensiveness of the search.

Conclusion

There is low-to-very low certainty evidence suggesting that therapeutic exercise may improve CVA, neck disability, and pain intensity in individuals with FHP defined by a CVA < 50°. Both neck-focused and combined neck – thoracic exercise programs showed short-term improvements in postural alignment, while combined programs appeared to produce more consistent effects over longer intervention periods and on pain intensity. However, due to methodological limitations, short follow-up periods, and heterogeneity across interventions and study populations, these findings should be interpreted with caution.

Author contributions

CRedit: **Andoni Carrasco-Uribarren:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Writing – original draft, Writing – review & editing; **Luis Ceballos-Laita:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing; **Silvia Pérez-Guillén:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing; **Sandra Jiménez-Del-Barrio:** Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing; **Daniel Pantaleón-Hernández:** Data curation, Formal analysis, Software, Writing – original draft, Writing – review & editing; **Sara Cabanillas-Barea:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Andoni Carrasco-Uribarren is a professor at Universitat Internacional de Catalunya (Spain) and coordinator of the Master of Science in Orthopedic Manual Physiotherapy. His primary area of research focuses on the management of cervical spine disorders and cervicogenic headaches. He has published numerous studies in high-impact journals regarding manual therapy and clinical reasoning, and he actively leads educational innovation projects aimed at transforming musculoskeletal training through active methodologies and the integration of advanced technology in the classroom.

Luis Ceballos-Laita is a professor at Universidad de Valladolid (Spain) and a prominent researcher in the field of Musculoskeletal Physiotherapy. He has published extensively in peer-reviewed journals and is actively involved in clinical trials aimed at evaluating the effectiveness of manual therapy and exercise, while also contributing to the advancement of evidence-based practice through his roles in postgraduate education and professional associations.

Silvia Pérez-Guillén is an associate professor at Universitat Internacional de Catalunya (Spain) and member of the

Standards Committee of IFOMPT. Her primary area of expertise lies in Manual and Musculoskeletal Physiotherapy, with a strong focus on educational innovation within the Health Sciences. She has spearheaded various projects involving active learning methodologies, such as the use of e-rubrics for the assessment of manual skills and team-based-learning strategies.

Sandra Jiménez-Del-Barrio is a professor at Universidad de Valladolid (Spain) and a prominent researcher in the field of Musculoskeletal Physiotherapy. Her primary area of expertise focuses on the management of musculoskeletal conditions and the efficacy of manual therapy interventions. She has a prolific research background in clinical reasoning and the assessment of movement dysfunction, having published numerous articles in high-impact journals. In addition to her research, she is actively involved in academic innovation, contributing to the development of evidence-based clinical practice and postgraduate specialized training.

Daniel Pantaleón-Hernández is an associate lecturer at Universitat Internacional de Catalunya (Spain). His primary area of expertise involves management of the cervical region. He also collaborates in the strategic implementation of active learning methodologies and the integration of digital infrastructure, to enhance student-centered learning. He has a strong background in pedagogical leadership, focusing on the quality and transformation of physiotherapy curricula through the development of professional competencies and the adoption of cutting-edge teaching tools.

Sara Cabanillas-Barea is an associate professor at Universitat Internacional de Catalunya (Spain) and researcher specialized in the field of Musculoskeletal Physiotherapy. Her primary area of expertise focuses on the assessment and management of orthopedic conditions, with a particular emphasis on the integration of manual therapy and therapeutic exercise in the cervical region. She has contributed to several research projects aimed at improving clinical outcomes in patients with musculoskeletal pain and is actively involved in academic innovation, focusing on the development of practical competencies and evidence-based clinical reasoning for physiotherapy students.

ORCID

Andoni Carrasco-Uribarren  <http://orcid.org/0000-0002-0906-9114>

Luis Ceballos-Laita  <http://orcid.org/0000-0002-2170-8783>

Silvia Pérez-Guillén  <http://orcid.org/0000-0003-1331-6656>

Sandra Jiménez-Del-Barrio  <http://orcid.org/0000-0002-7460-3061>

Daniel Pantaleón-Hernández  <http://orcid.org/0009-0002-9029-8527>

Sara Cabanillas-Barea  <http://orcid.org/0000-0002-3755-5646>

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